

# Disproof of TLMC Conjecture 00000000404

## Schur coefficients of $s_{(2)}[s_{(n)}]$ at even-indexed $\nu$

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### Abstract

Conjecture 00000000404 asserts that in the Schur expansion of the plethysm  $s_{(2)}[s_{(n)}]$ , the coefficients indexed by even partitions  $\nu$  are all even. We exhibit the counterexample  $n = 2$ :

$$s_{(2)}[s_{(2)}] = s_{(4)} + s_{(2,2)},$$

whose two nonzero Schur coefficients are both 1, hence odd. Under every natural reading of “even-indexed  $\nu$ ” ( $|\nu|$  even, all parts even, first part even, or even length) this violates the conjecture. The identity is derived in the power-sum basis, cross-checked by two independent Weyl dimension counts, and fully machine-verified in core Lean with zero axioms.

## 1 The conjecture

**00000000404.** *Definition:* Schur plethysm  $s_\lambda \circ s_\mu$ . *Conjecture:* among the Schur expansion coefficients of  $s_{(2)}[s_{(n)}]$ , the coefficients of even-indexed  $\nu$  are all even.

The phrase “even-indexed  $\nu$ ” is ambiguous; we refute it under all four natural readings simultaneously (Section 3).

## 2 The attack at $n = 2$

Recall the  $\lambda$ -ring identity for the second symmetric power: for any symmetric function  $f$ ,

$$s_2[f] = \tfrac{1}{2}(f \cdot f + \psi^2(f)), \quad \psi^2(p_k) = p_{2k},$$

since  $s_2 = (p_1^2 + p_2)/2$  and plethysm is a composition of  $\Lambda$ -ring endomorphisms. With  $f = s_2 = (p_{11} + p_2)/2$  we get

$$s_2 \cdot s_2 = \frac{(p_{11} + p_2)^2}{4} = \frac{p_{1111} + 2p_{211} + p_{22}}{4}, \quad \psi^2(s_2) = \frac{p_{22} + p_4}{2},$$

hence

$$s_{(2)}[s_{(2)}] = \frac{p_{1111} + 2p_{211} + 3p_{22} + 2p_4}{8}. \tag{1}$$

(The subscripts record part multiplicities, e.g.  $p_{211} = p_2 p_1^2$ .)

Schur functions expand as  $s_\nu = \sum_\rho \chi^\nu(\rho) p_\rho / z_\rho$ , where  $\chi^\nu$  is the irreducible character of  $S_{|\nu|}$  at cycle type  $\rho$  and  $z_\rho = \prod_i i^{m_i} m_i!$ . By Hall orthogonality  $\langle p_\rho, s_\nu \rangle = \chi^\nu(\rho)$ , the Schur coefficients of (1) are  $c_\nu = \sum_\rho a_\rho \chi^\nu(\rho)$  with  $a = (1, 2, 0, 3, 2)/8$  over  $(p_{1111}, p_{211}, p_{31}, p_{22}, p_4)$ . The  $S_4$  character table (columns =  $1^4, 21^2, 31, 22, 4$ ; verified by the Murnaghan–Nakayama rule in `reproduce.py`) is:

| $\nu \backslash \rho$ | 1111 | 211 | 31 | 22 | 4  |
|-----------------------|------|-----|----|----|----|
| (4)                   | 1    | 1   | 1  | 1  | 1  |
| (31)                  | 3    | 1   | 0  | -1 | -1 |
| (22)                  | 2    | 0   | -1 | 2  | 0  |
| (211)                 | 3    | -1  | 0  | -1 | 1  |
| (1111)                | 1    | -1  | 1  | 1  | -1 |

Dotting  $a$  with each row:

$$\begin{aligned}
8c_{(4)} &= 1 + 2 + 0 + 3 + 2 = 8, & 8c_{(31)} &= 3 + 2 + 0 + 0 + 0 = 0, \\
8c_{(22)} &= 2 + 0 + 0 + 6 + 0 = 8, & 8c_{(211)} &= 3 - 2 + 0 - 3 + 2 = 0, \\
& & 8c_{(1111)} &= 1 - 2 + 0 + 3 - 2 = 0.
\end{aligned}$$

Therefore

$$s_{(2)}[s_{(2)}] = s_{(4)} + s_{(2,2)}, \quad c_{(4)} = 1, \quad c_{(2,2)} = 1 \quad (2)$$

and both nonzero coefficients are **odd**.

## 2.1 Dimension cross-check

For  $d = 4$ :  $\dim \text{Sym}^2(\text{Sym}^2 \mathbb{C}^4) = \binom{10+1}{2} = 55$ , while  $\dim S_{(4)} \mathbb{C}^4 = \binom{4+4-1}{4} = 35$  and  $\dim S_{(2,2)} \mathbb{C}^4 = 20$  (hook-content / Weyl formula). The multiplicity-free decomposition (2) matches the count  $55 = 35 + 20$ . For  $d = 2$  likewise  $\dim \text{Sym}^2(\text{Sym}^2 \mathbb{C}^2) = \binom{4}{2} = 6 = 5 + 1$  ( $\dim s_{(4)} = 5$ ,  $\dim s_{(2,2)} = 1$ ), again consistent.

## 3 Every reading of “even-indexed” is refuted

| Reading of “even-indexed $\nu$ ” | Witness $\nu$ at $n = 2$ | Coefficient |
|----------------------------------|--------------------------|-------------|
| $ \nu $ even                     | (4) and (2, 2)           | 1 (odd)     |
| all parts even                   | (4) and (2, 2)           | 1 (odd)     |
| first part even                  | (4) and (2, 2)           | 1 (odd)     |
| length $\ell(\nu)$ even          | (2, 2)                   | 1 (odd)     |

In every case the coefficient is  $1 \not\equiv 0 \pmod{2}$ . The conjecture is false.

## 4 Boundary and scope

- $n = 1$ :  $s_{(2)}[s_{(1)}] = s_{(2)}$  with coefficient 1 (odd);  $\nu = (2)$  has even size, all parts even, first part even. The failure is not an  $n = 2$  accident.
- **General  $n$  (computed for  $n \leq 6$ ):** the expansion is multiplicity-free,  $s_{(2)}[s_{(n)}] = \sum_{j=0}^{\lfloor n/2 \rfloor} s_{(2n-2j, 2j)}$ , every coefficient being exactly 1; each summand has even size, all parts even and first part even, and for  $n \geq 2$  also even length. So no choice of  $n$  rescues the conjecture.

## 5 Machine verification

The directory `lean4/` contains a core-Lean (no Mathlib) formalization (`lakefile.toml`, `lean-toolchain` `leanprover/lean4:v4.33.1`):

- `attack_expansion`: identity (1), proved by `rfl` on the explicit  $\lambda$ -ring computation;
- `character_table_orthogonal`: the  $S_4$  table above is column-orthogonal (weights 1, 6, 8, 3, 6), all 25 pairs checked by `decide`;
- `conjecture_404_false` / `parity_odd`: the five coefficients of (2) and their parity, by `decide`;
- `dim_check`:  $55 = 35 + 20$ ; `boundary_n1`: the  $n = 1$  case.

The axiom audit (`lake env lean Check.lean`) reports that all six theorems *do not depend on any axioms*: no `sorry`, no `propext`, no `Quot.sound`, no `Classical.choice`, no `Lean.ofReduceBool`.

## 6 Reproducing

```
python3 reproduce.py                # 13 checks, exit 0
cd lean4 && lake build && lake env lean Check.lean
tectonic main.tex --outdir build    # this PDF
```

All paths are relative to this submission directory; no external packages are required beyond the Python standard library and a Lean toolchain.